

RoadSOS Mobile Application

Complete Technical Architecture & System Documentation Specification

Executive Overview: RoadSOS is a state-of-the-art emergency assistance and intelligent road safety platform built for Android devices. The system features multi-sensor fusion crash detection, sigmoid severity scoring, ride mode controls, offline-first NoSQL database synchronization, and automated location-aware emergency SMS dispatch.

1. Core System Architecture & Modules

Module 1: User Management & Authentication

- **Driver Registration & Authentication:** Supports secure session creation, driver profile management, and role-based permissions (User / Admin).
- **Offline Session Persistence:** Integrates local Isar database session storage to allow drivers to authenticate and access emergency functions even in offline environments without internet connectivity.
- **Cloud Profile Sync:** Syncs profile details with backend REST API endpoints whenever internet access is restored.

Module 2: Emergency Contacts Management

- **Local Emergency Contacts Storage:** Allows drivers to add, edit, and store primary emergency contacts locally on-device.
- **Native Address Book Integration:** Integrates with native Android contacts (via flutter_contacts) to quickly import family members or medical contacts.
- **Zero-Latency Readiness:** Ensures contact phone numbers are cached in memory for instant emergency SMS dispatch.

Module 3: Crash Detection, Ride Mode & Sensor Fusion Engine

- **Continuous Ride Mode:** A dedicated toggle mode that activates live 6-axis IMU hardware background monitoring when the user is driving.
- **Multi-Stage Noise Filtering:** Includes consecutive high-G acceleration filters, rotational velocity orientation analyzers, and post-impact driver inactivity detectors to prevent false alarms.
- **Mathematical Sigmoid Severity Scoring:** Computes weighted multi-variable severity scores (0.0 to 1.0) mapping to **CRITICAL**, **HIGH**, **MEDIUM**, and **LOW** severity categories.
- **Live Telemetry & Metric Testing:** Features expandable telemetry logs detailing peak G-forces, rotation speeds, and driver inactivity, plus live phone sensor sampling.

2. Complete Technical Stack

Category	Technologies & Libraries Used	Key Role / Purpose
Mobile Framework	Flutter (Dart SDK >=3.0.0 <4.0.0)	Cross-platform native mobile app framework
State Management	Flutter Riverpod (v2.4.9) & ChangeNotifier	Compile-safe reactive state management & stream listeners

Hardware Sensors	Sensors Plus (v7.1.0)	Real-time 6-axis IMU (UserAccelerometer & Gyroscope)
Local Databases	Isar NoSQL (v3.1.0), Hive (v2.2.3), SharedPreferences	High-speed offline database for users, contacts & crash logs
Networking	HTTP Client (v1.1.2), Connectivity Plus (v6.0.0)	REST API client & automatic online/offline state monitor
Native Telephony	URL Launcher (v6.2.4), Flutter Contacts (v1.1.7)	Native SMS URI scheme dispatch & device address book
Target OS / SDK	Android 15 (API 35/36, compileSdk = 36)	Android mobile operating system target

3. Severity Scoring Algorithm & Operations Routing

The severity score S is computed mathematically using four weighted sigmoid curves:
Score = $0.40 \cdot S_{\text{impact}} + 0.25 \cdot S_{\text{orient}} + 0.20 \cdot S_{\text{inactive}} + 0.15 \cdot S_{\text{speed}}$
where each component evaluates sigmoid response: $S(x) = 1 / (1 + \exp(-k \cdot (x - m)))$.

Severity Level	Score Threshold	Sensor Criteria	Automated Operation Executed
CRITICAL	Score ≥ 0.70	High G-force ($> 4.0G$) + Spin/Rollover	Instant Emergency Alert Modal + Immediate Automated SMS Dispatch with GPS location to contacts.
HIGH	$0.50 \leq \text{Score} < 0.70$	Significant Impact ($2.8G - 4.0G$)	Instant Emergency Alert Modal + Immediate Automated SMS Dispatch to contacts.
MEDIUM	$0.30 \leq \text{Score} < 0.50$	Moderate Impact ($1.8G - 2.8G$)	60-Second Countdown Check Dialog. Driver can tap 'I'm Okay' to cancel; dispatches SMS if un-cancelled.
LOW	Score < 0.30	Minor Bump / Pot-hole ($< 1.8G$)	In-App Minor Safety Notification Banner (No emergency SMS sent).

4. Recommended Machine Learning (TFLite) Integration

In addition to the physics-based sigmoid engine, RoadSOS supports on-device **TensorFlow Lite (TFLite)** 1D CNN / CNN-LSTM deep learning model execution:

- **Model Architecture:** 1D Convolutional Neural Network with LSTM recurrent time-sequence layers.
- **Input Tensor:** [1, 200, 6] representing 200 time-series samples recorded at 50Hz across 6 sensor features (Accel X,Y,Z + Gyro X,Y,Z).
- **Output Tensor:** [1, 4] Softmax classification probabilities across [NORMAL, LOW, MEDIUM, CRITICAL].
- **Flutter Plugin:** Executed via tflite_flutter plugin.

5. Deployment & Build Specifications

- **Build Artifact:** Release Android APK (app-release.apk, size ~73.3 MB).
- **USB Auto-Deployment Command:** flutter run --release -d <device-id>
- **Manual Installation:** Transfer app-release.apk to phone storage and tap to install.
- **Target Devices:** Android 8.0+ up to Android 15 (Samsung Galaxy SM A146B API 35 tested).